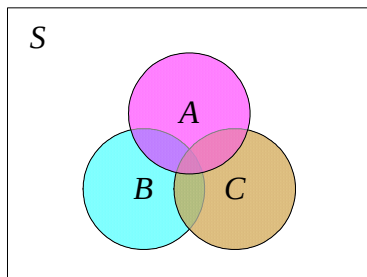


Chapter 3 Inclusion-Exclusion Principle

- Sets A and B are subset of S , the number of objects that belong neither to A nor to B :

$$|S \setminus A \setminus B| = |S| - |A| - |B| + |A \cap B|$$

$$\Pr\{A \cup B\} = \Pr\{A\} + \Pr\{B\} - \Pr\{A \cap B\}$$



$$|A \cup B \cup C| = (|A| + |B| + |C|) - (|A \cap B| + |B \cap C| + |A \cap C|) + |A \cap B \cap C|$$

3.1 Probabilistic Inclusion-Exclusion Principle

- Let $A_1, \dots, A_n$ be n events which may be intersecting and dependent.
 P_r denotes the probability that exactly r events occurs in a random trial.

$$P_r = \sum_{1 \leq j_1 < \dots < j_r \leq n} \Pr\{A_{j_1} \cdots A_{j_r} \bar{A}_{j_{r+1}} \cdots \bar{A}_{j_n}\},$$

where $j_1, \dots, j_n$ is a permutation of $(1, 2, \dots, n)$

- Denote that $J_r = \{(j_1, \dots, j_r) : 1 \leq j_1 < \dots < j_r \leq n\}$, and define

$$\begin{aligned} S_k &:= \sum_{J_k} \Pr\{A_{j_1} \cdots A_{j_k}\} \\ &= \sum_{J_k} \Pr\{A_{j_1} \cdots A_{j_k} (A_{j_{k+1}} \cup \bar{A}_{j_{k+1}}) \cdots (A_{j_n} \cup \bar{A}_{j_n})\} \\ &= \sum_{r=k}^n \binom{r}{k} \sum_{J_r} \Pr\{A_{j_1} \cdots A_{j_r} \bar{A}_{j_{r+1}} \cdots \bar{A}_{j_n}\} \\ &= \sum_{r=k}^n \binom{r}{k} P_r \end{aligned}$$

- But we are interested in P_r rather than S_k . We need to invert the above.

$$\begin{aligned}\sum_{k=r}^n (-1)^{r+k} \binom{k}{r} S_k &= \sum_{k=r}^n \sum_{s=k}^n (-1)^{r+k} \binom{k}{r} \binom{s}{k} P_s \\ &= \sum_{s=r}^n \left(\sum_{k=r}^s (-1)^{k-r} \binom{k}{r} \binom{s}{k} \right) P_s\end{aligned}$$

However, as can be easily verified,

$$\sum_{k=r}^s (-1)^{k-r} \binom{k}{r} \binom{s}{k} = \sum_{l=0}^{s-r} (-1)^l \binom{s-r}{l} = \begin{cases} 1 & \text{if } r = s \\ 0 & \text{otherwise} \end{cases}$$

We conclude that:

- **Theorem 3.1 (Inclusion-Exclusion Principle)**

Let $A_1, \dots, A_n$ be events in an arbitrary probability space, and let P_r be the probability of exactly r of them to occur. Then

$$P_r = \sum_{k=r}^n (-1)^{r+k} \binom{k}{r} S_k$$

● **Example 3.1 (Computing a Distribution Through Its Moments)**

Let X be a random variable defined on $\{0, 1, \dots, n\}$ and let $E_k[X] = E[X(X-1)\dots(X-k+1)]$ be the k th factorial moment of X . We shall prove that

$$\Pr\{X = r\} = \frac{1}{r!} \sum_{k=r}^n (-1)^{r+k} \frac{E_k[X]}{(k-r)!}$$

Proof: Let $A_i = \{X \geq i\}$, for all $i = 1, \dots, n$. We now evaluate $S_k = \sum_{J_k} \Pr\{\bigcap_{j \in J_k} A_j\}$

$$S_1 = \sum_{j=1}^n \Pr\{X \geq j\} = \sum_{i=1}^n i \Pr\{X = i\} = E[X]$$

$$\begin{aligned} S_2 &= \sum_{i=1}^n \sum_{j=i+1}^n \Pr\{X \geq j\} = \sum_{j=2}^n (j-1) \Pr\{X \geq j\} \\ &= \sum_{k=2}^n \Pr\{X = k\} \sum_{j=1}^{k-1} j = \frac{E_2[X]}{2} \end{aligned}$$

$$S_k = \sum_{1 \leq j_1 < \dots < j_k \leq n} \Pr\{X \geq j_1, \dots, X \geq j_k\} = \frac{1}{k!} E_k[X]$$

But $\{X = r\}$ is equivalent to the event that exactly r of A_i occur, since $\{X = r\} =$

$$A_1 A_2 \cdots A_r \bar{A}_{r+1} \cdots \bar{A}_n = P_r$$

$$\begin{aligned}\Pr\{X = r\} &= P_r = \sum_{k=r}^n (-1)^{r+k} \binom{k}{r} S_k \\ &= \frac{1}{r!} \sum_{k=r}^n (-1)^{r+k} \frac{E_k[X]}{(k-r)!}\end{aligned}$$

● **Example 3.2 (Laplace's Formula on the Sum of I.I.D. Uniform Distributions)**

Let $X_1, \dots, X_n$ be independent random variables uniformly distributed on the interval $(0,1)$. We shall prove Laplace's formula

$$\Pr\{X_1 + \dots + X_n < x\} = \frac{1}{n!} \sum_{k=0}^n (-1)^k \binom{n}{k} (x-k)_+^n$$

where $x_+ = \max\{0, x\}$. Consider the volume of the following two sets

$$\Omega = \{(x_1, \dots, x_n) : x_1 + \dots + x_n < x, 0 \leq x_i \leq 1, i = 1, \dots, n\}$$

$$\tilde{\Omega} = \{(x_1, \dots, x_n) : x_1 + \dots + x_n < x, x_i \geq 0, i = 1, \dots, n\}$$

and define $A_i = \tilde{\Omega} \cap \{x_i > 1\}$. When none of A_i occurs, we can set $r = 0$ and obtain

$$P_r = \sum_{k=r}^n (-1)^{r+k} \binom{k}{r} S_k$$

$$\Pr\{X_1 + \dots + X_n < x\} = P_0 = \sum_{k=0}^n (-1)^k S_k$$

where $S_k = \sum_{1 \leq j_1 < \dots < j_k \leq n} V(A_{j_1}, \dots, A_{j_k})$

To compute $V(A_{j_1}, \dots, A_{j_k})$ we must set k variables in the equation $x_1 + \dots + x_n = x$

to 1; hence

$$V(A_{j_1}, \dots, A_{j_k}) = \frac{(x-k)_+^n}{n!}$$

$$\begin{aligned} \Pr\{X_1 + \dots + X_n < x\} &= \sum_{k=0}^n (-1)^k S_k \\ &= \sum_{k=0}^n (-1)^k \sum_{1 \leq j_1 < \dots < j_k \leq n} V(A_{j_1}, \dots, A_{j_k}) \\ &= \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{(x-k)_+^n}{n!} \\ &= \frac{1}{n!} \sum_{k=0}^n (-1)^k \binom{n}{k} (x-k)_+^n \end{aligned}$$

- How to estimate the probability that at least one event occurs?
- **Corollary 3.2**

For arbitrary events the following holds:

$$\Pr\{\bigcup_{i=1}^n A_i\} = \sum_{k=1}^n (-1)^{k+1} \sum_{J_k} \Pr\{\bigcap_{j \in J_k} A_j\} = \sum_{k=1}^n (-1)^{k+1} S_k$$

Proof:

$$\begin{aligned} \Pr\{\bigcup_{i=1}^n A_i\} &= \sum_{r=1}^n P_r \\ &= \sum_{r=1}^n \sum_{k=r}^n (-1)^{r+k} \binom{k}{r} S_k = \sum_{k=1}^n (-1)^k S_k \sum_{r=1}^k \binom{k}{r} (-1)^r \\ &= \sum_{k=1}^n (-1)^k S_k [(1-1)^k - 1] = \sum_{k=1}^n (-1)^{k+1} S_k \end{aligned}$$

Note that $(a+b)^k = \binom{k}{0}a^k + \binom{k}{1}a^{k-1}b^1 + \binom{k}{2}a^{k-2}b^2 + \dots + \binom{k}{k-1}a^1b^{k-1} + \binom{k}{k}b^k$, and

we set $a = 1$ and $b = -1$.

